

Course Introduction

COMPUTER NETWORKS

CAN THO UNIVERSITY

09/2025

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Course Objectives

- Fundamental concepts of computer networks
- Issues related to data transmission in computer networks
- Layered design principles in computer networks
- Roles and Functions of network components
- Commonly used protocols in computer networks

Learning Outcomes (1)

- Describe the general structure of a computer network
- Explain the benefits of computer networks
- Classify different types of networks
- Explain concepts related to components in the software architecture of computer networks, such as protocols, services, and interfaces

Learning Outcomes (2)

- List the names (in Vietnamese/English) of the OSI model layers in the correct order
- Describe the roles and functions of each OSI layer
- Explain the operation principles of common protocols corresponding to each OSI layer
- Master the English terminology related to computer networking

Theory Content

- Overview of computer networks
- Components of a computer network
- OSI model
- Physical layer
- Data link layer
- LAN & MAC
- Network layer
- Transport layer
- Application layer

Practical Content

- Practice environment: Docker, Kathara, Wireshark, tcpdump
- Topics
 - Network simulation with Kathara
 - Packet analysis with Wireshark and tcpdump
 - Layer 2 internetworking
 - Static routing
 - Dynamic routing algorithms
 - Application-layer protocols

Course Structure and Assessment

- Theory: 30 periods
- Practice: 30 periods
- Evaluation:
 - Mid-term tests: 20%
 - Lab assessment: 30%
 - Final exam: 50%

Main References

- Computer Networks Textbook, [Ngo Ba Hung, Pham The Phi], Can Tho University, 2014
- Computer Networks Lab Lectures [2017]
- Slides in both English and Vietnamese